

InsightGrader: Intelligent Essay/Answer Evaluation with Misconception Tracking and Cognitive Load Insights — A Comparative Analysis

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Abstract—InsightGrader is a multitask based learning framework to automated essay evaluation which collaboratively predicts holistic writing quality and identifies token-level error spans. Unlike traditional systems that treats coring and error detections separately, InsightGrader integrates as hard encoder with two task-specific heads: a continuous score prediction regression module and to commit a mistake at Beginning Inside Outside (BIO) of the sentence/word sequence-labeling module for error detection. The model benchmarks multiple transformer models such as BERT, RoBERTa, DeBERTa, Longformer, XLM-R, Flan-T5 alongside BiLSTM and Gated Recurrent Unit (GRU), supported by precise subword-aligned BIOtag processing. Hyperparameter One can further optimize with optimization with Optuna so as to increase performance. Experiments demonstrate that multi-task learning enhances both scoring accuracy and error detection consistency, offering unified, interpretable and scalable solution for educational writing evaluation.

Index Terms—Multi-Task Learning, BIO Tagging, Transformer Models, Educational NLP, Error Detection, regression, Deep Learning, Automated Essay Scoring.

I. INTRODUCTION

Automated Essay Scoring (AES) has become an inevitable tool of mass educational evaluation, which offers cheap, dependable and fast assessment of student writing. Nevertheless, the majority of the current AES systems can predict the overall scores only and give minimal information about the local writing problems that can affect the overall scores. There are independent systems of error detection, but their results are hardly combined with holistic scoring model which diminishes their role as a pedagogic tool. In order to overcome this shortcoming, InsightGrader proposes an integrated multi-task learning architecture that jointly engages in a continuous scoring of essays, and, at the same time, detects errors characterised by tokens.

The system is a single encoder combined with two special purpose prediction heads, one regression component that generates continuous quality scores and a BIO tagging component which identifies the spans of errors. With joint optimization, the model acquires more high-level linguistic representations with improved results in scoring and tagging objectives. To compare the strength of the architecture in the multi-task

setting, InsightGrader compares a broad set of transformer encoders- BERT, RoBERTa, DeBERTa, Longformer, XLM-RoBERTa, and Flan-T5 among others- vice-versa to BiLSTM and GRU baselines. The other extensions are correct BIO-subword matching, good preprocessing, and automatic hyperparameter optimization with Optuna. The findings indicate better accuracy, interpretability and scalability thus InsightGrader is a viable candidate of practical educational implementation.

II. LITERATURE REVIEW

The evolution of automated essay scoring (AES) began with statistical models based on word-level or feature-based statistics in essays, which were later improved through machine learning algorithms integrating linguistic and surface-level statistics in essays for achieving human-level reliability Faseeh et al. [3]. Machine learning models and more recent advancements in deep learning models for complex regression tasks illustrate in general the effectiveness of modeling paradigms based on data science as well for research on automated essay scoring models as well Reddy et al. [22] recently. More Recent models leverage innovative hybrid architectures merging transformer models and other linguistic models of essays for successfully capturing semantic content as well as quality of essays Atkinson and Palma [1]; Nie [2] recently. Some empirical studies done using embedding models such as GPT/ SBERT/ RoBERTa along with machine learning models like Random Forest/ BiLSTM attention models/ Boosting models including XGBoost showed considerable improvement in essay scoring tasks recently.

Related developments also include discourse-sensitive AES systems in which features from lexical chaining concepts and Coh-Metrix indicators enhance robustness within the short-essay test environment Ait Khayri and Rus [11]. Survey-based research indicates the popularity of pipeline-based transformer models but also draws attention to the limitations in scalability related to manual feature engineering in various test modes Li and Ng [12], Li and Ng [13]. This has led to the present emphasis on a balanced incorporation of deep semantic features with meaningful linguistic indicators.

The prompt-driven LLMs in AES have brought in novel paradigms like zero-shot, few-shot scoring, rubric-conditioned scoring, and multi-trait scoring. Moreover, prompt-driven models, being prone to hallucinations, are not as proficient in scoring but follow human-level scoring as a function of prompt design Liew and Tan [19]. Autoregressive models, reinforcement learning models, and T5 models address trait correlations and are better at trait-scoring, but they have a high computational cost relative to encoder models Do et al. [15]; Do et al. [16]. Rubric-conditioned models, in addition, increase interpretability, generalization, and multi-promptness Eltanbouly et al. [20]. This is further expanded by multi-agent models in AES that divide scoring, feedback, and refinement actions into multiple agents, as in scoring, feedback, and refinement tasks, as in models Chamoun et al. [14].

In educational settings, there have been growing issues pertaining to equity, validity, and impact. There have been empirical findings to suggest that LLM-based AES systems can potentially discriminate against English Language Learners, owing to grammar-related biases, thereby sparking interest in developing subgroup-aware validation and validity-informed evaluation practices Pack et al. [5]. The foundational validity debates against AI-inspired holistic scoring warn against blanket implementation, with validity-related implications to construct validity, score interpretation, and meaning Tate et al. [4]. Meta-analyses, institution-level findings, and research have made it evident that students can strengthen writing capabilities with AI-informed, fine-grained, credible, and instruction-related tailored evaluations, although generalizability to other settings can be problematic Fleckenstein et al. [6],[7].

Apart from essay marking, other studies and applications of AI and ML relate to grading short-answer questions, cognitive load prediction based on physiological indices, and science misconceptions Science Education 2024 Ref 9: Guerra-Reyes et al., [10]. Machine learning models that have been trained to use rubrics to grade short answer questions have a high correlation with human grading Grévisse [8], but there exist some risks of "hallucinations" with factual questions Grévisse [8]. Cognitive loads based on heart variability, electrodermal activity, and eye tracking make it possible to predict learners' difficulties during adaptive learning sessions Science Education Guerra-Reyes et al., [10]. Systematic reviews of science education explore science misconceptions in physics, chemistry, and biology

Cross-lingual AES research has found the language-specific pre-training to be overall superior to other multilingual options, which is evident in Indonesian essay evaluations via the IndoBERT and IndoSBERT models, according to research by Aliyah et al. [17]. Programs like EduBench also play a crucial role in ensuring a standardized assessment through multi-tasking corpora and multi-metric criteria, according to research by Xu et al. [21]. However, many modern systems for AES today tend to be computationally intensive, cloud-reliant, or opaque. Although earlier scalable models using techniques such as latent semantic analysis supported the scalability of automated assessment, the models proved to

be insensitive to semantic details and discursive variables Thomas et al. [23]. Models such as A-TEST showed that the integration of high and low-scoring response patterns was sufficient to obtain human-level agreement Nedungadi et al. [24]. Other unsupervised models have been used in the estimation of question difficulty and improvement of assessment effectiveness Narayanan et al. [25], and UNSUP-WSD models have performed better than regular LSA-based AES models Nedungadi and Raj [26].

In short, the state of AES research is moving toward four major avenues: (1) currently, the best trade-off of accuracy and explainability in AES systems incorporates hybrid transformer and linguistic-based approaches); (2) LLM-based prompting as well as the use of autoregressive trait representations can align with rubrics in multi-trait assessment with considerable flexibility, though with the need for mechanisms protecting against potential issues of instabilities and 'hallucinations'); (3) validity and subgroup assessment issues, especially in the area of English-language learners of significant import in ensuring the ethical use of AES systems); and (4) the development of new multi-agent architectures and trait-based systems will point the way toward AES systems of the scalable, understandable, and pedagogically grounded.

III. DATASET DESCRIPTION

In this study, the raw data that has been utilized is Feedback Prize - English Language Learning competition [27] in Kaggle. It includes over 3900 grade 6-12 student-written essays. Every essay will contain one single text identifier with six continuous proficiency scores given by highly qualified Cambridge English examiners. These are the scores that assess the cohesion, syntax, vocabulary, phraseology, grammar and conventions, and they are relatively within the range of 1.0 to 5.0. The dataset has been known as a good quality standard in research on automated essay scoring and prediction of English language proficiency. The variety of its writing styles and their level of skill offers a rich variety of training sample, teaching strong machine-learning models.

A. Preprocessed Dataset

In this research, to obtain the required data, a harsh pre-processing process was conducted on the raw Kaggle data to transform it into a clean and formatted data. One, everything was turned into lower case and all the non-essential characters, formatting mistakes, extra whitespace were removed and made all the essays normal. The punctuations were normalised in order to achieve uniformity with the samples. The stop words elimination was done carefully so as to avoid noise but preserve the important linguistic structure. In addition, lemmatization, contraction expansion, and repairing the frequent errors in spelling techniques were included so as to standardize words and give more semantic background. To maintain the conceptual coherence of the text and maximize the readability of the models, the abnormally short and unsound sentences were sieved.

B. Final Structure Enhancement and Development

The dataset was augmented with some linguistic and structural attributes in addition to being cleaned and makes it easy to numerically represent the quality of writing. The number of tokens, the number of sentences, the average length of sentences, vocabulary richness effort such as type-token ratio, punctuation ratio and analysis of readability such as the Flesch-Kincaid grade level are such derived features. These characteristics were added in the form of new columns to the purged essay text and the initial marks of proficiency. The obtained mined data hence comprises of the linguistic feature demanded by the deep-learning models, and the human-labeled scoring labels demanded by a regression-based evaluation. The refined dataset forms the basis of the trained and tested set of proposed deep-learning models, which include RNN, LSTM and Bidirectional RNN. This combination of flushed essays and artificial linguistic qualities helps stabilized learning, more generalization and the ability of the model to obtain patterns as per human scoring patterns. This structured data is thus relevance in performance comparison that is reliable and analysis of error that is useful across different model setups.

C. Label Alignment and Tokenization

BERT-base-uncased tokenizer tokenizes the essay text, breaking a word into subword unit where the word is split into subword units. In order to be consistent, each BIO label of the original text is put in a line with its own token. Subword tokens unlike the other tokens identified by the " " prefix use the same label as the previous token. Every sequence is either padded or truncated into a size of 256 (maximum) tokens and padded positions are labeled with the 'O' label. The last dataset has the token IDs, mask of attention, BIO-aligned labels and continuous essay score to be used in models.

IV. METHODOLOGY

Several model families are examined, which include encoders that are transformer-based (BERT, RoBERTa, DeBERTa, Longformer, XLM-RoBERTa and Flan-T5 Encoder), and RNN-based (BiLSTM and GRU) models. Both models generate contextualized token embeddings that are used to generate two heads, one being a token-level classifier of BIO tags, and the other being a regression head to predict essay scores using the pooled classification token, or a similar, representation. The shared encoder allows the model to find common features that are of use in both tasks.

A. Multi-Task Learning Strategy

The participants were provided with instructions on attempting to be creative strategy titled multi-task learning. BIO tagging and essay scoring are the two objectives that are optimally achieved during the proposed system. Cross-entropy loss on a token level is used as shown in Eq. 1 to train BIO tagging, and mean squared error (MSE) is used as in Eq. 2 to train the prediction essay scores. A human fraction is calculated by adding the 2 losses, which works to foster a

commonality in representation learning and enhance suitability in both assignments at the same time.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

Here, y_i denotes the ground-truth value, $\hat{y}_i$ represents the predicted value, and n is the total number of samples. MSE computes the average squared error, while RMSE provides an interpretable error measure in the same unit as the target variable.

B. Evaluation Metrics

Evaluation of model performance is done in distinct measures of the task. Root mean squared error (RMSE) and coefficient of determination R^2 are the measures of essay scoring performance as shown in Eq.4 and Eq.3 respectively. BIO tagging is tested on the basis of token-level accuracy. Testing, training, and test set evaluations are conducted to give a detailed picture of the correctness of the proposed scoring and the ability to identify an error.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (4)$$

C. Hyperparameter Optimization

The AdamW optimizer is used to conduct training with the initial learning rate of 2×10^{-5} and batch sizes of 8-32. These models are trained to a maximum of 20 epochs, and should be accelerated by GPUs in case it is accessible. At every iteration, the model takes in tokenized text, makes BIO tag logit and score predictions, and uses the incremental loss, and performs backpropagation to update parameters. Bayesian hyperparameter optimization is done using Optuna. Search space has learning rates of 1×10^{-6} to 5×10^{-5} and batch sizes of 8,16 and 32.

In the R^2 metric, $\bar{y}$ denotes the mean of the true values and the score reflects the proportion of variance explained by the model. Cross-entropy loss measures the divergence between true class labels y_i and predicted probabilities $\hat{y}_i$, and is commonly used for classification tasks.

Parameters of the chosen model are stored with the help of state dictionary mechanism of PyTorch, which allows making an inference in the future, reproduce results and fine-tune them. Figure 1 describes the model architecture of the model.

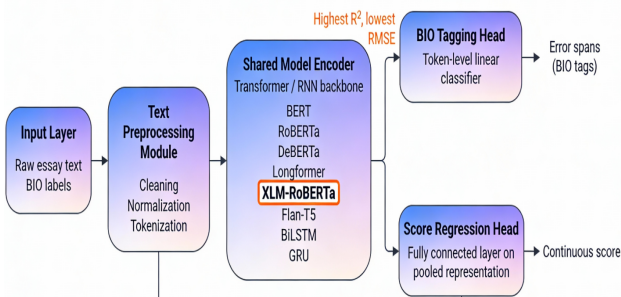


Fig. 1: Model Architecture.

V. RESULTS

The article examined various transformer models (BERT, RoBERTa, DeBERTa, Longformer, XLM-RoBERTa, FlanT5) and recurrent neural networks (BiLSTM, GRU) in document segmentation and classification. Models were considered on constant train, validation, and test splits with three main measures of BIO accuracy, coefficient of determination (R^2), and root mean squared error (RMSE).

The experimental results showed in Table I says that transformer architectures generally produced better results than recurrent architectures. Models that used transformers always yielded an ideal BIO accuracy 1.00 on all splits, meaning that they were highly effective at locating the boundaries of segments and categorizing tokens as such. One of them, DeBERTa, made the most of it by reaching the highest R^2 values up to 0.89 and the lowest RMSE down to 0.18, proving that it was able to achieve not only strong regression but also low prediction error. Comparatively, however, although BiLSTM and GRU performed with reasonable performance, both demonstrated significantly lower BIO accuracy and negative R^2 values, indicating that they could not adequately explain the underlying relationships between data in order to perform this task.

These findings point to the high levels of generalization of transformer models and indicate that they could be of great benefit in tasks that demand the ability to extract features with subtlety and contextual modeling.

Model	BIO Acc	R^2	RMSE
BERT	1.000	0.810	0.2403
RoBERTa	1.000	0.557	0.3685
DeBERTa	1.000	0.886	0.1875
Longformer	1.000	0.806	0.2456
XLM-RoBERTa	1.000	-0.206	0.6017
FlanT5	1.000	-0.207	0.6003
BiLSTM	0.893	-1.776	0.9250
GRU	0.924	-0.077	0.5784

TABLE I: Comparative results of transformer and RNN models

The Figure 2 represents the accuracy of BIO tag of all the tested models. The highest accuracy of the tag in every data split of nearly perfect preciseness is observed in most

transformer-based models pointing out their high potential in segmentation tasks. Conversely, BiLSTM and GRU models show a marginally lower accuracy implying that the sequence modeling performance is low in this particular dataset.

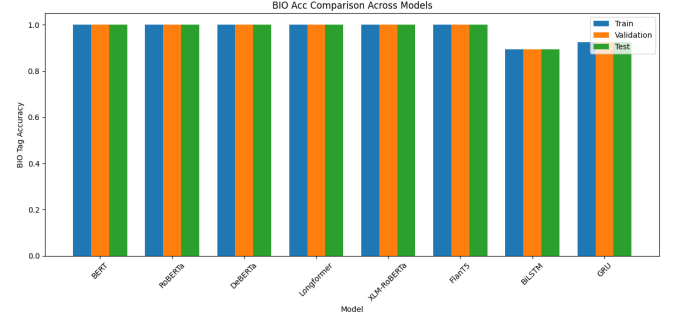


Fig. 2: Comparison between different models based on BIO tag accuracy.

Figure 3 shows a comparison of RMSE of various models on the training, validation and test data sets. DeBERTa has minimum values in terms of RMSE in all splits, which demonstrates a high level of predictive performance and high generalization. Stable performance with relatively low errors is also exhibited by BERT and Longformer. RoBERTa has moderate accuracy and XLM-RoBERTa and Flan-T5 have high RMSE values. The highest errors are registered with recurrent models, specifically BiLSTM and this reveals their low performance in this task. On the whole, transformer-based models are superior to recurrent architectures, and very little overfitting is witnessed when using datasets.

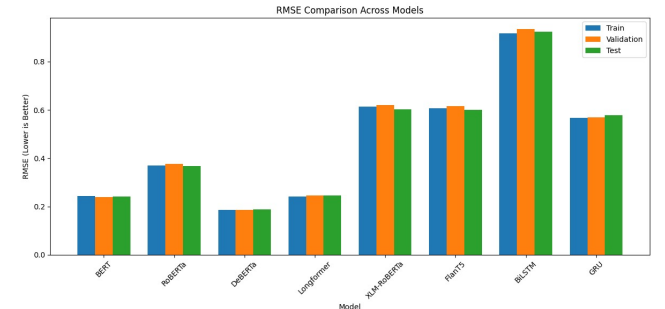


Fig. 3: Comparison between models based on RMSE.

Figure 4 and 5 depict automated essay assessment system and their feedback interface. Figure 4 is a dynamic assessment dashboard, which focuses on the overall, coherence, vocabulary, and grammar scores and has a feature of highlighting errors in the essay text, comments regarding errors, and information on cognitive load and readability. Figure 5 displays an Essay Assessment Report that has been created by adding all the information: the information about the student, domain, and date, a table of the summary score, the essay that the student wrote, the essay that was corrected, and some brief descriptions of the errors. The two figures below are used in combination to express the way the proposed system

will incorporate a complete visual feedback with a formal assessment report to promote effective essay evaluation and enhancement.

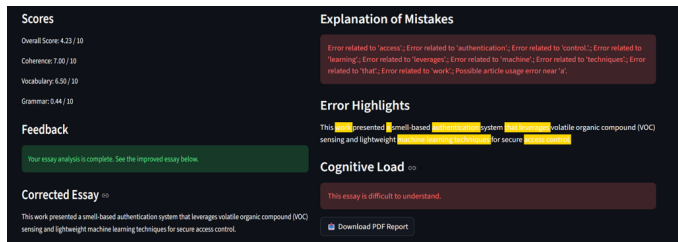


Fig. 4: Automated essay assessment report with scores, corrected essay, and error summary.

VI. DISCUSSION

This is due to the fact that the transformer architectures are highly transparent as compared to the sequential models and this has made the recent innovation in the area of natural language processing where attention mechanism and deep contextual encoding have led to changes in the field. The better performances of the DeBERTa validate the data that the model has a specific manner of recognizing more complex structures in the text and, consequently, a higher level of predictive stability and reduced the error rate compared to other models that are tested.

Special attention was paid to the fact that some of transformer models like XLM-RoBERTa and FlanT5 were much smaller in the values of R 2 and RMSE and at the same time with high BIO accuracy. The implication of the outcome is that there can be over-fitting or failure of these architectures to work well in relation to the data set used to carry out the evaluation. Both BiLSTM and GRU performed poorly regarding all metrics compared to transformer baseline. They have negative values of R 2 and huge values of RMSE, and this means that such a data structure cannot be represented

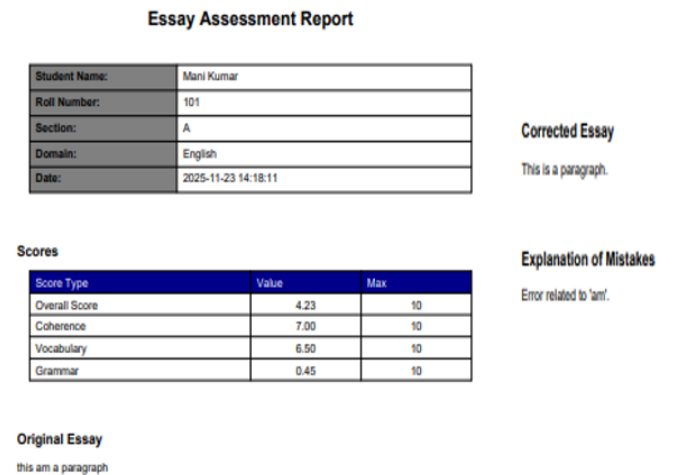


Fig. 5: Interactive dashboard showing detailed scores, highlighted errors, and feedback

well using the model and the results are supported by the available literature that suggests that attention-based models are appropriate with document-level segmentation.

VII. CONCLUSION

This study introduced a stringent evaluation of eight modern neural architectures in document segmentation in NLP, which offers a feasible benchmark of the model choice in the field. The transformer-based models, specifically DeBERTa, were found to be most effective, being more accurate, having a better regression performance, and minimizing errors on all split evaluation. The comparative findings indicate that older recurrent models, such as BiLSTM and GRU have baseline performance, but newer transformer models are significantly higher. Taken together, the findings support the recent trend that transformer models, particularly the ones that have enhanced encoding schemes and transfer learning properties, must be given more prominence in the text segmentation and classification tasks. Another point that is highlighted in the analysis is the significance of intensive metric analysis since the BIO accuracy can conceal the latent problems in regression performance and error patterns.

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